

Integration by Substitution: Complete Tutorial with 20 Practice Problems

Calculus Tutorial

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1 Introduction to Integration by Substitution

1.1 What is Integration by Substitution?

Integration by substitution is used when an integral contains a **function and its derivative**. It's essentially the **reverse of the chain rule** from differentiation.

1.2 The Basic Idea

Think of it like this:

Replace a complicated part of the integral with a simpler variable to make integration easier.

2 The Substitution Formula

The method is based on this idea:

If you have:

$$\int f(g(x)) \cdot g'(x) dx$$

Then:

1. Let $u = g(x)$ 2. Then $du = g'(x) dx$ 3. Substitute to get: $\int f(u) du$ 4. Integrate with respect to u 5. Substitute back $u = g(x)$

3 Simple Step-by-Step Example

3.1 Example 1: Basic Polynomial

$$\int 2x(x^2 + 1)^3 dx$$

Solution:

$$\text{Let } u = x^2 + 1$$

$$\text{Then } du = 2x dx$$

$$\text{Substitute: } \int (x^2 + 1)^3 \cdot 2x dx = \int u^3 du$$

$$\text{Integrate: } = \frac{u^4}{4} + C$$

$$\text{Substitute back: } = \frac{(x^2 + 1)^4}{4} + C$$

4 When to Use Substitution

Look for these patterns:

- **Power of a bracket:** $(3x + 2)^5$, $(x^2 - 1)^{3/2}$
- **Function inside another function:** $\sin(4x)$, $\cos(x^2)$
- **Fraction where numerator is derivative of denominator:** $\frac{2x}{x^2+3}$
- **Exponential with composite argument:** e^{5x} , e^{x^2}
- **Square root of a function:** $\sqrt{x+1}$, $\sqrt{3x-2}$

5 The Substitution Process in Detail

5.1 Step 1: Identify u

Choose u to be the "inside function" – usually what's in parentheses, under a square root, or in an exponent.

5.2 Step 2: Find du

Differentiate u with respect to x : $du = \frac{du}{dx} dx$

5.3 Step 3: Substitute

Replace all x terms with u terms. Sometimes you need to solve for dx or adjust constants.

5.4 Step 4: Integrate

Integrate with respect to u .

5.5 Step 5: Substitute Back

Replace u with the original x expression.

6 Special Cases

6.1 When dx Doesn't Match Perfectly

Sometimes du is off by a constant factor. For example:

$$\int x(x^2 + 1)^3 dx$$

Here $u = x^2 + 1$, $du = 2x dx$, but we only have $x dx$. So:

$$x dx = \frac{1}{2} du$$

6.2 Definite Integrals with Substitution

For definite integrals, you can either:

1. Substitute back to x before evaluating, OR
2. Change the limits to u -values and evaluate with u

7 20 Practice Problems with Solutions

7.1 Easy Level

1. $\int 2x(x^2 + 5) dx$

Solution:

$$\text{Let } u = x^2 + 5$$

$$du = 2x \, dx$$

$$\begin{aligned}\int 2x(x^2 + 5) \, dx &= \int u \, du \\ &= \frac{u^2}{2} + C \\ &= \frac{(x^2 + 5)^2}{2} + C\end{aligned}$$

2. $\int \cos(3x) \, dx$

Solution:

$$\text{Let } u = 3x$$

$$du = 3 \, dx \quad \Rightarrow \quad dx = \frac{1}{3} du$$

$$\begin{aligned}\int \cos(3x) \, dx &= \int \cos u \cdot \frac{1}{3} du \\ &= \frac{1}{3} \sin u + C \\ &= \frac{1}{3} \sin(3x) + C\end{aligned}$$

3. $\int e^{4x} \, dx$

Solution:

$$\text{Let } u = 4x$$

$$du = 4 \, dx \quad \Rightarrow \quad dx = \frac{1}{4} du$$

$$\begin{aligned}\int e^{4x} \, dx &= \int e^u \cdot \frac{1}{4} du \\ &= \frac{1}{4} e^u + C \\ &= \frac{1}{4} e^{4x} + C\end{aligned}$$

4. $\int \frac{2x}{x^2 + 1} \, dx$

Solution:

$$\text{Let } u = x^2 + 1$$

$$du = 2x \, dx$$

$$\begin{aligned}\int \frac{2x}{x^2 + 1} \, dx &= \int \frac{1}{u} \, du \\ &= \ln |u| + C \\ &= \ln(x^2 + 1) + C \quad (\text{since } x^2 + 1 > 0)\end{aligned}$$

5. $\int \sin(5x) \, dx$

Solution:

$$\text{Let } u = 5x$$

$$du = 5 \, dx \quad \Rightarrow \quad dx = \frac{1}{5} du$$

$$\begin{aligned} \int \sin(5x) \, dx &= \int \sin u \cdot \frac{1}{5} du \\ &= -\frac{1}{5} \cos u + C \\ &= -\frac{1}{5} \cos(5x) + C \end{aligned}$$

7.2 Medium Level

6. $\int (3x+1)^4 \, dx$

Solution:

$$\text{Let } u = 3x + 1$$

$$du = 3 \, dx \quad \Rightarrow \quad dx = \frac{1}{3} du$$

$$\begin{aligned} \int (3x+1)^4 \, dx &= \int u^4 \cdot \frac{1}{3} du \\ &= \frac{1}{3} \cdot \frac{u^5}{5} + C \\ &= \frac{(3x+1)^5}{15} + C \end{aligned}$$

7. $\int \frac{1}{\sqrt{x+4}} \, dx$

Solution:

$$\text{Let } u = x + 4$$

$$du = dx$$

$$\begin{aligned} \int \frac{1}{\sqrt{x+4}} \, dx &= \int u^{-1/2} \, du \\ &= 2u^{1/2} + C \\ &= 2\sqrt{x+4} + C \end{aligned}$$

8. $\int x\sqrt{x^2+9} \, dx$

Solution:

$$\text{Let } u = x^2 + 9$$

$$du = 2x \, dx \quad \Rightarrow \quad x \, dx = \frac{1}{2} du$$

$$\begin{aligned} \int x\sqrt{x^2+9} \, dx &= \int \sqrt{u} \cdot \frac{1}{2} du \\ &= \frac{1}{2} \int u^{1/2} \, du \\ &= \frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C \\ &= \frac{1}{3} (x^2+9)^{3/2} + C \end{aligned}$$

9. $\int \tan(2x) dx$

Solution:

Let $u = 2x$

$$du = 2 dx \quad \Rightarrow \quad dx = \frac{1}{2} du$$

$$\begin{aligned} \int \tan(2x) dx &= \int \tan u \cdot \frac{1}{2} du \\ &= \frac{1}{2} \int \tan u du \\ &= \frac{1}{2} (-\ln |\cos u|) + C \\ &= -\frac{1}{2} \ln |\cos(2x)| + C \end{aligned}$$

10. $\int \frac{1}{x \ln x} dx$

Solution:

Let $u = \ln x$

$$du = \frac{1}{x} dx$$

$$\begin{aligned} \int \frac{1}{x \ln x} dx &= \int \frac{1}{u} du \\ &= \ln |u| + C \\ &= \ln |\ln x| + C \end{aligned}$$

7.3 Advanced Level

11. $\int x \sqrt{1-x^2} dx$

Solution:

Let $u = 1 - x^2$

$$du = -2x dx \quad \Rightarrow \quad x dx = -\frac{1}{2} du$$

$$\begin{aligned} \int x \sqrt{1-x^2} dx &= \int \sqrt{u} \cdot \left(-\frac{1}{2}\right) du \\ &= -\frac{1}{2} \int u^{1/2} du \\ &= -\frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C \\ &= -\frac{1}{3} (1-x^2)^{3/2} + C \end{aligned}$$

12. $\int 2xe^{x^2} dx$

Solution:

$$\begin{aligned}\text{Let } u &= x^2 \\ du &= 2x \, dx \\ \int 2xe^{x^2} \, dx &= \int e^u \, du \\ &= e^u + C \\ &= e^{x^2} + C\end{aligned}$$

13. $\int \frac{x}{1+x^2} \, dx$

Solution:

$$\begin{aligned}\text{Let } u &= 1+x^2 \\ du &= 2x \, dx \quad \Rightarrow \quad x \, dx = \frac{1}{2} du \\ \int \frac{x}{1+x^2} \, dx &= \int \frac{1}{u} \cdot \frac{1}{2} du \\ &= \frac{1}{2} \ln |u| + C \\ &= \frac{1}{2} \ln(1+x^2) + C\end{aligned}$$

14. $\int 2x \sin(x^2) \, dx$

Solution:

$$\begin{aligned}\text{Let } u &= x^2 \\ du &= 2x \, dx \\ \int 2x \sin(x^2) \, dx &= \int \sin u \, du \\ &= -\cos u + C \\ &= -\cos(x^2) + C\end{aligned}$$

15. $\int \frac{1}{(5x+1)^2} \, dx$

Solution:

$$\begin{aligned}\text{Let } u &= 5x+1 \\ du &= 5 \, dx \quad \Rightarrow \quad dx = \frac{1}{5} du \\ \int \frac{1}{(5x+1)^2} \, dx &= \int \frac{1}{u^2} \cdot \frac{1}{5} du \\ &= \frac{1}{5} \int u^{-2} \, du \\ &= \frac{1}{5} \cdot \frac{u^{-1}}{-1} + C \\ &= -\frac{1}{5(5x+1)} + C\end{aligned}$$

7.4 Tricky/Exam Type

16. $\int \frac{2x}{\sqrt{x^2+4}} dx$

Solution:

$$\text{Let } u = x^2 + 4$$

$$du = 2x dx$$

$$\begin{aligned}\int \frac{2x}{\sqrt{x^2+4}} dx &= \int \frac{1}{\sqrt{u}} du \\ &= 2\sqrt{u} + C \\ &= 2\sqrt{x^2+4} + C\end{aligned}$$

17. $\int \frac{3x^2}{x^3+1} dx$

Solution:

$$\text{Let } u = x^3 + 1$$

$$du = 3x^2 dx$$

$$\begin{aligned}\int \frac{3x^2}{x^3+1} dx &= \int \frac{1}{u} du \\ &= \ln|u| + C \\ &= \ln(x^3+1) + C\end{aligned}$$

18. $\int \frac{\cos(\sqrt{x})}{2\sqrt{x}} dx$

Solution:

$$\text{Let } u = \sqrt{x} = x^{1/2}$$

$$du = \frac{1}{2}x^{-1/2} dx = \frac{1}{2\sqrt{x}} dx$$

$$\begin{aligned}\int \frac{\cos(\sqrt{x})}{2\sqrt{x}} dx &= \int \cos u du \\ &= \sin u + C \\ &= \sin(\sqrt{x}) + C\end{aligned}$$

19. $\int \frac{1}{(x+1)\ln(x+1)} dx$

Solution:

$$\text{Let } u = \ln(x+1)$$

$$du = \frac{1}{x+1} dx$$

$$\begin{aligned}\int \frac{1}{(x+1)\ln(x+1)} dx &= \int \frac{1}{u} du \\ &= \ln|u| + C \\ &= \ln|\ln(x+1)| + C\end{aligned}$$

20. $\int x e^{x^2} dx$

Solution:

$$\text{Let } u = x^2$$

$$du = 2x dx \Rightarrow x dx = \frac{1}{2} du$$

$$\begin{aligned} \int x e^{x^2} dx &= \int e^u \cdot \frac{1}{2} du \\ &= \frac{1}{2} e^u + C \\ &= \frac{1}{2} e^{x^2} + C \end{aligned}$$

8 Quick Recognition Guide

8.1 The One-Second Rule

If you see a function AND its derivative in the integrand $\rightarrow$ USE SUBSTITUTION!

8.2 Common Patterns to Recognize

Pattern	Choose u	Example
$\int f(ax+b) dx$	$u = ax+b$	$\int (3x+2)^4 dx$
$\int f(x^n) \cdot x^{n-1} dx$	$u = x^n$	$\int x^2 e^{x^3} dx$
$\int \frac{f'(x)}{f(x)} dx$	$u = f(x)$	$\int \frac{2x}{x^2+1} dx$
$\int f(\sqrt{x}) \cdot \frac{1}{\sqrt{x}} dx$	$u = \sqrt{x}$	$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$
$\int f(e^x) \cdot e^x dx$	$u = e^x$	$\int e^x \cos(e^x) dx$

9 Tips and Tricks

9.1 What if du isn't exactly there?

Sometimes the derivative is off by a constant. Just adjust:

$$\text{If } du = k \cdot (\text{what you have}), \text{ then multiply by } \frac{1}{k}$$

9.2 Definite Integrals

Two approaches:

1. Integrate with substitution, then substitute back to x , then use original limits.
2. Change limits: if $u = g(x)$, then when $x = a$, $u = g(a)$; when $x = b$, $u = g(b)$.

9.3 Multiple Substitutions

Sometimes you need to substitute more than once, especially with nested functions.

10 Comparison with Integration by Parts

Integration by Substitution	Integration by Parts
Based on chain rule Use when you see function & its derivative u = inner function Usually simpler	Based on product rule Use when you see product of two functions u = function that simplifies when differentiated Often requires multiple applications

11 Practice Makes Perfect

Remember:

- Practice recognizing when to use substitution
- Always check your answer by differentiating
- Look for the function-derivative pair
- When in doubt, try a substitution and see if it simplifies things